

Design and Implementation of a Database Management System for Hostel Accommodation and Student Services Management System

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Problem Identification

Existing Problems



Record Keeping

Manual hostel records are difficult to maintain and student transfer history may be lost

Bed Allocation

Duplicate bed allocation may occur and room capacity may be exceeded

Tracking & Monitoring

Difficult to track vacant and occupied beds; fee payments and dues are difficult to monitor

Services & Requests

Complaints and maintenance requests may be missed; visitor records require systematic management

Aim & Objectives

Aim

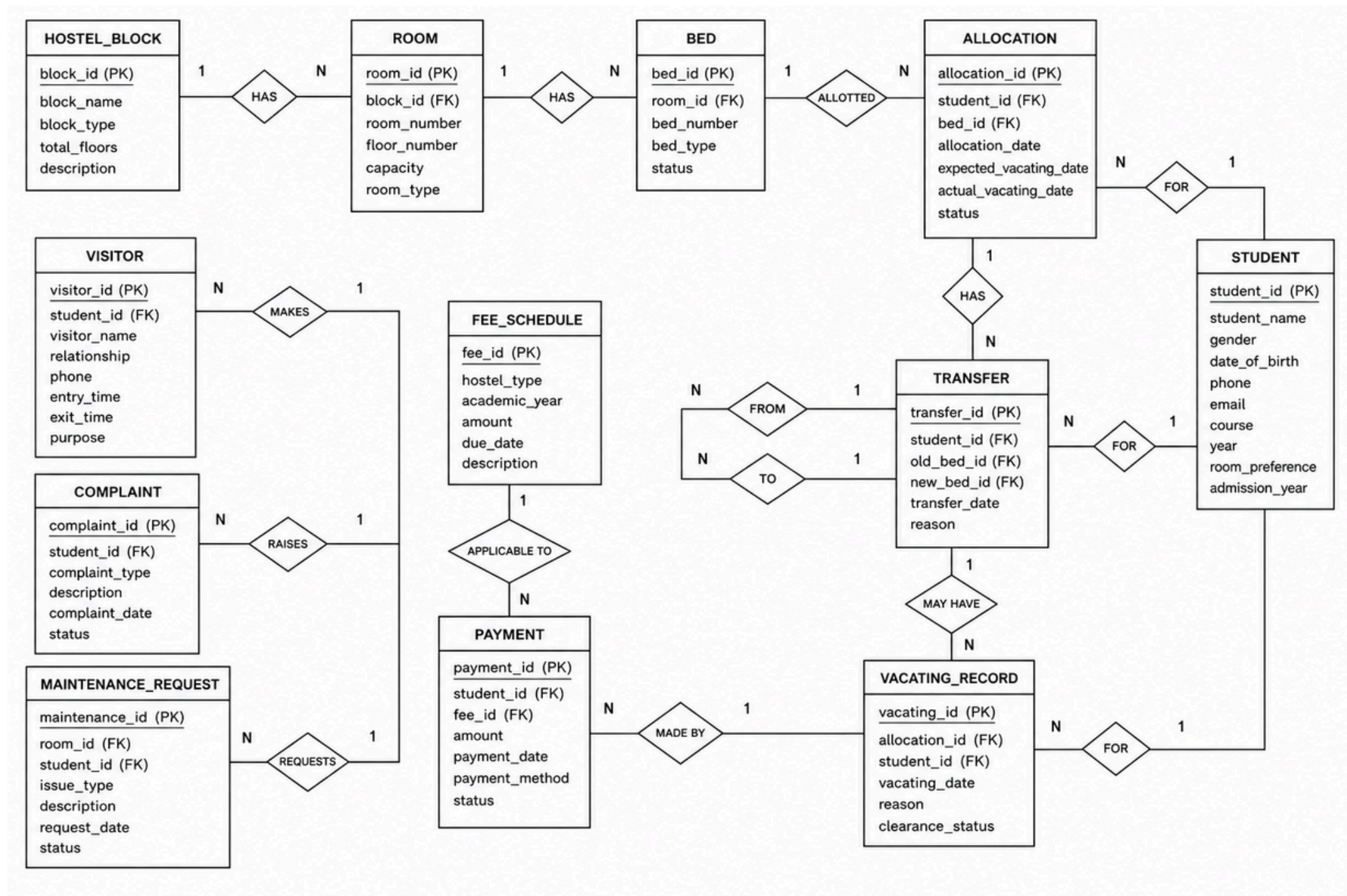
To design and implement a relational database system for efficient management of hostel accommodation and student services.



Objectives

- Manage student and hostel information
- Manage blocks, rooms and beds
- Prevent duplicate bed allocation
- Track student transfers and vacating
- Manage hostel fees and payments
- Maintain visitor records
- Track complaints and maintenance
- Generate occupancy and vacancy reports
- Maintain data integrity and consistency

ER DIAGRAM:



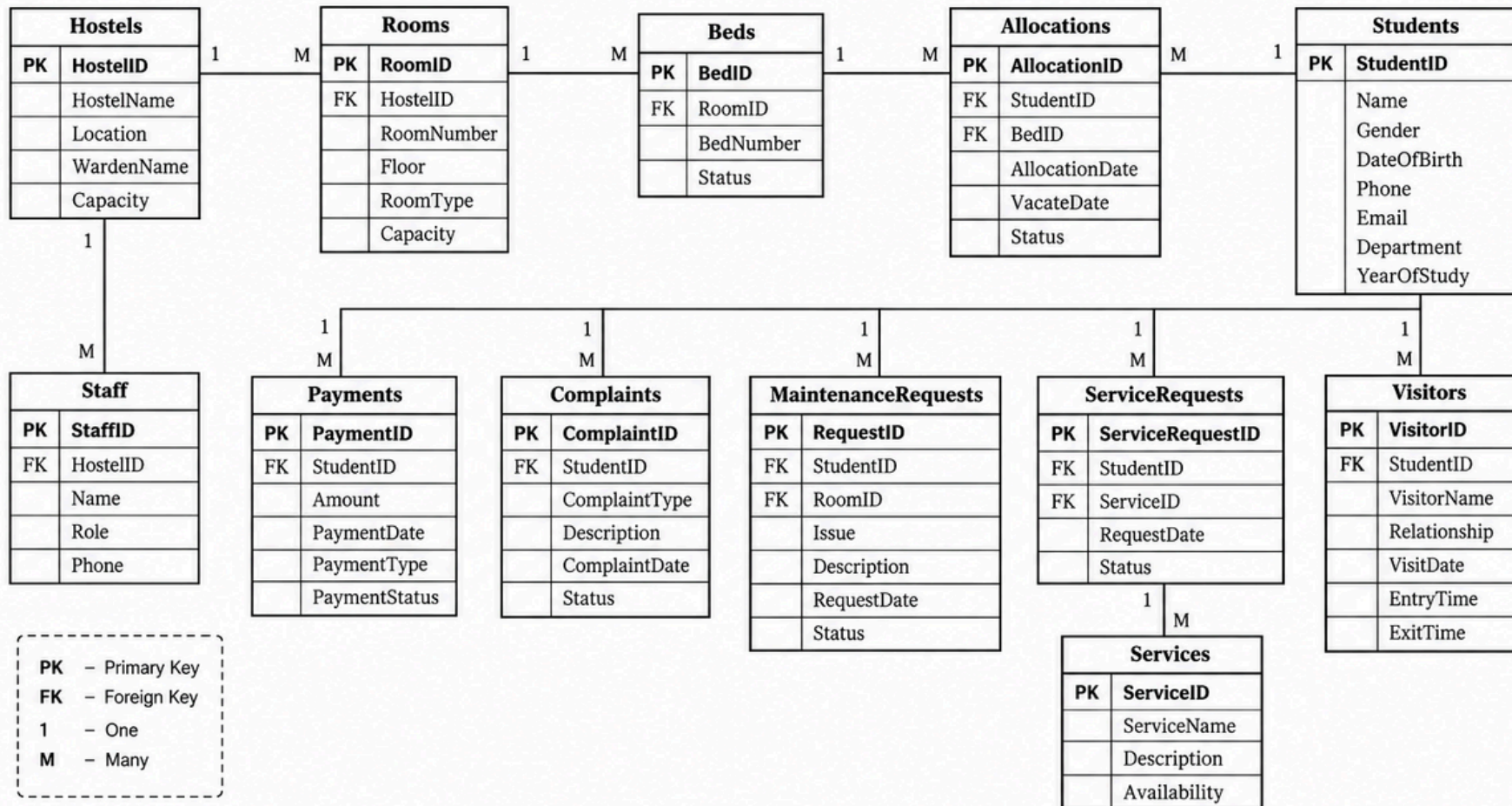
Initial Relational Schema

Proposed Tables

#	Table Name	Key Attributes
1	HOSTEL_BLOCK	block_id (PK)
2	ROOM	room_id (PK), block_id (FK)
3	BED	bed_id (PK), room_id (FK)
4	STUDENT	student_id (PK)
5	ALLOCATION	allocation_id (PK), student_id (FK), bed_id (FK)
6	TRANSFER	transfer_id (PK), student_id (FK)
7	FEE_SCHEDULE	fee_id (PK)
8	PAYMENT	payment_id (PK), student_id (FK), fee_id (FK)
9	VISITOR	visitor_id (PK), student_id (FK)
10	COMPLAINT	complaint_id (PK), student_id (FK)
11	MAINTENANCE_REQUEST	maintenance_id (PK), student_id (FK), room_id (FK)
12	VACATING_RECORD	vacating_id (PK), allocation_id (FK), student_id (FK)

Database Design

The database is designed using a relational model. It consists of 12 tables that store information about students, hostels, rooms, beds, accommodation, payments, complaints, maintenance requests, services, staff and visitors. The tables are related using primary keys (PK) and foreign keys (FK) to maintain data consistency and reduce redundancy.



Integrity Constraints

	PRIMARY KEY Unique identification of records
	FOREIGN KEY Maintains table relationships
	NOT NULL Prevents missing required values
	UNIQUE Prevents duplicate values
	CHECK Validates data values
	DEFAULT Provides default values

Rules

One active resident per bed	One active allocation per student
Room capacity cannot be exceeded	Allocation dates must be valid
Vacating date must be valid	Payment amount must be positive
Payment cannot exceed outstanding fees	

MYSQL CODE TO CREATE TABLE :

```
mysql> CREATE DATABASE HostelDB;
Query OK, 1 row affected (0.03 sec)

mysql> USE HostelDB;
Database changed
mysql> CREATE TABLE Students (
  -> StudentID INT PRIMARY KEY AUTO_INCREMENT,
  -> Name VARCHAR(100) NOT NULL,
  -> Gender VARCHAR(10),
  -> DateOfBirth DATE,
  -> Phone VARCHAR(15),
  -> Email VARCHAR(100) UNIQUE,
  -> Department VARCHAR(100),
  -> YearOfStudy INT
  -> );
Query OK, 0 rows affected (0.08 sec)

mysql>
mysql> CREATE TABLE Hostels (
  -> HostelID INT PRIMARY KEY AUTO_INCREMENT,
  -> HostelName VARCHAR(100) NOT NULL,
  -> Location VARCHAR(150),
  -> WardenName VARCHAR(100),
  -> Capacity INT
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> CREATE TABLE Rooms (
  -> RoomID INT PRIMARY KEY AUTO_INCREMENT,
  -> HostelID INT,
  -> RoomNumber VARCHAR(20) NOT NULL,
  -> Floor INT,
  -> RoomType VARCHAR(30),
  -> Capacity INT,
  -> FOREIGN KEY (HostelID) REFERENCES Hostels(HostelID)
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> CREATE TABLE Beds (
  -> BedID INT PRIMARY KEY AUTO_INCREMENT,
  -> RoomID INT,
  -> BedNumber VARCHAR(20),
  -> Status VARCHAR(20) DEFAULT 'Available',
  -> FOREIGN KEY (RoomID) REFERENCES Rooms(RoomID)
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> CREATE TABLE Allocations (
  -> AllocationID INT PRIMARY KEY AUTO_INCREMENT,
  -> StudentID INT,
  -> BedID INT,
  -> AllocationDate DATE NOT NULL,
  -> VacateDate DATE,
  -> Status VARCHAR(20) DEFAULT 'Active',
  -> FOREIGN KEY (StudentID) REFERENCES Students(StudentID),
  -> FOREIGN KEY (BedID) REFERENCES Beds(BedID)
  -> );
Query OK, 0 rows affected (0.05 sec)
```

```
mysql>
mysql> CREATE TABLE Payments (
  -> PaymentID INT PRIMARY KEY AUTO_INCREMENT,
  -> StudentID INT,
  -> Amount DECIMAL(10,2),
  -> PaymentDate DATE,
  -> PaymentType VARCHAR(50),
  -> PaymentStatus VARCHAR(20),
  -> FOREIGN KEY (StudentID) REFERENCES Students(StudentID)
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> CREATE TABLE Complaints (
  -> ComplaintID INT PRIMARY KEY AUTO_INCREMENT,
  -> StudentID INT,
  -> ComplaintType VARCHAR(100),
  -> Description TEXT,
  -> ComplaintDate DATE,
  -> Status VARCHAR(30) DEFAULT 'Pending',
  -> FOREIGN KEY (StudentID) REFERENCES Students(StudentID)
  -> );
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> CREATE TABLE MaintenanceRequests (
  -> RequestID INT PRIMARY KEY AUTO_INCREMENT,
  -> StudentID INT,
  -> RoomID INT,
  -> Issue VARCHAR(150),
  -> Description TEXT,
  -> RequestDate DATE,
  -> Status VARCHAR(30) DEFAULT 'Pending',
  -> FOREIGN KEY (StudentID) REFERENCES Students(StudentID),
  -> FOREIGN KEY (RoomID) REFERENCES Rooms(RoomID)
  -> );
Query OK, 0 rows affected (0.07 sec)

mysql>
mysql> CREATE TABLE Services (
  -> ServiceID INT PRIMARY KEY AUTO_INCREMENT,
  -> ServiceName VARCHAR(100),
  -> Description TEXT,
  -> Availability VARCHAR(30)
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> CREATE TABLE ServiceRequests (
  -> ServiceRequestID INT PRIMARY KEY AUTO_INCREMENT,
  -> StudentID INT,
  -> ServiceID INT,
  -> RequestDate DATE,
  -> Status VARCHAR(30) DEFAULT 'Requested',
  -> FOREIGN KEY (StudentID) REFERENCES Students(StudentID),
  -> FOREIGN KEY (ServiceID) REFERENCES Services(ServiceID)
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> CREATE TABLE Staff (
  -> StaffID INT PRIMARY KEY AUTO_INCREMENT,
  -> Name VARCHAR(100),
  -> Role VARCHAR(50),
  -> Phone VARCHAR(15),
  -> HostelID INT,
  -> FOREIGN KEY (HostelID) REFERENCES Hostels(HostelID)
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> CREATE TABLE Visitors (
  -> VisitorID INT PRIMARY KEY AUTO_INCREMENT,
  -> StudentID INT,
  -> VisitorName VARCHAR(100),
  -> Relationship VARCHAR(50),
  -> VisitDate DATE,
  -> EntryTime TIME,
  -> ExitTime TIME,
  -> FOREIGN KEY (StudentID) REFERENCES Students(StudentID)
  -> );
Query OK, 0 rows affected (0.05 sec)
```

tables:

```
mysql> show tables;
Tables_in_campushosteladb
allocations
beds
complaints
hostels
maintenancerequests
payments
rooms
servicerequests
services
staff
students
visitors
12 rows in set (0.00 sec)
```

```
mysql>
mysql> SELECT * FROM Students;
```

StudentID	Name	Gender	DateOfBirth	Phone	Email	Department	YearOfStudy
1	Rahul Kumar	Male	2005-04-12	9876543210	rahul@gmail.com	CSE	2
2	Ananya Sharma	Female	2004-08-20	9876543211	ananya@gmail.com	ECE	3
3	Arjun Reddy	Male	2005-01-15	9876543212	arjun@gmail.com	CSE	2
4	Priya Singh	Female	2006-03-10	9876543213	priya@gmail.com	EEE	1
5	Karan Mehta	Male	2004-11-25	9876543214	karan@gmail.com	MECH	3
6	Sneha Rao	Female	2005-06-18	9876543215	sneha@gmail.com	CSE	2
7	Vikram Patel	Male	2004-09-22	9876543216	vikram@gmail.com	ECE	3
8	Meera Nair	Female	2005-12-05	9876543217	meera@gmail.com	CIVIL	2

8 rows in set (0.00 sec)

```
mysql>
mysql> SELECT * FROM Hostels;
```

HostelID	HostelName	Location	WardenName	Capacity
1	Block A	North Campus	Mr. Rajesh	200
2	Block B	South Campus	Mrs. Priya	150
3	Block C	East Campus	Mr. Suresh	180

3 rows in set (0.00 sec)

```
mysql>
mysql> SELECT * FROM Rooms;
```

RoomID	HostelID	RoomNumber	Floor	RoomType	Capacity
1	1	A101	1	Double	2
2	1	A102	1	Triple	3
3	1	A201	2	Single	1
4	1	B101	1	Double	2
5	2	B102	1	Triple	3
6	2	B201	2	Single	1
7	3	C101	1	Double	2
8	3	C201	2	Triple	3

8 rows in set (0.00 sec)

```
mysql>
mysql> SELECT * FROM Beds;
```

BedID	RoomID	BedNumber	Status
1	1	A101-1	Occupied
2	1	A101-2	Available
3	2	A102-1	Occupied
4	2	A102-2	Occupied
5	2	A102-3	Available
6	3	A201-1	Available
7	4	B101-1	Occupied
8	4	B101-2	Available
9	5	B102-1	Occupied
10	5	B102-2	Available
11	5	B102-3	Available
12	6	B201-1	Occupied
13	7	C101-1	Available
14	7	C101-2	Available
15	8	C201-1	Available
16	8	C201-2	Available
17	8	C201-3	Available

17 rows in set (0.00 sec)

```
mysql>
mysql> SELECT * FROM Allocations;
```

AllocationID	StudentID	BedID	AllocationDate	VacateDate	Status
1	1	1	2026-06-10	NULL	Active
2	2	3	2026-06-11	NULL	Active
3	3	4	2026-06-12	NULL	Active
4	4	7	2026-06-13	NULL	Active
5	5	9	2026-06-14	NULL	Active
6	6	12	2026-06-15	NULL	Active

6 rows in set (0.00 sec)

```
mysql>
mysql> SELECT * FROM Payments;
```

PaymentID	StudentID	Amount	PaymentDate	PaymentType	PaymentStatus
1	1	45000.00	2026-06-10	Hostel Fee	Paid
2	2	45000.00	2026-06-11	Hostel Fee	Paid
3	3	45000.00	2026-06-12	Hostel Fee	Pending
4	4	40000.00	2026-06-13	Hostel Fee	Paid
5	5	40000.00	2026-06-14	Hostel Fee	Pending
6	6	45000.00	2026-06-15	Hostel Fee	Paid
7	7	45000.00	2026-06-16	Hostel Fee	Paid
8	8	40000.00	2026-06-17	Hostel Fee	Pending

8 rows in set (0.00 sec)

```
mysql>
mysql> SELECT * FROM Complaints;
```

ComplaintID	StudentID	ComplaintType	Description	ComplaintDate	Status
1	1	Internet	Wi-Fi is not working properly.	2026-07-01	Pending
2	2	Cleanliness	Room requires cleaning.	2026-07-02	Resolved
3	3	Water Supply	No water supply in bathroom.	2026-07-03	In Progress
4	4	Electricity	Room light is not working.	2026-07-04	Resolved
5	5	Food	Food quality needs improvement.	2026-07-05	Pending
6	6	Internet	Internet connection is slow.	2026-07-06	In Progress
7	7	Furniture	Chair is damaged.	2026-07-07	Pending
8	8	Cleanliness	Common area needs cleaning.	2026-07-08	Resolved

8 rows in set (0.00 sec)

```
mysql>
mysql> SELECT * FROM MaintenanceRequests;
```

RequestID	StudentID	RoomID	Issue	Description	RequestDate	Status
1	1	1	Fan Repair	Ceiling fan is making noise.	2026-07-06	Pending
2	2	2	Plumbing	Washbasin tap is leaking.	2026-07-07	In Progress
3	3	2	Electrical	Tube light is damaged.	2026-07-08	Resolved
4	4	4	Door Repair	Room door lock is damaged.	2026-07-09	Pending
5	5	5	Furniture	Study table needs repair.	2026-07-10	Resolved
6	6	1	Fan Repair	Fan is not working.	2026-07-11	Completed
7	7	4	Plumbing	Bathroom pipe is leaking.	2026-07-12	Pending
8	8	7	Electrical	Power socket is damaged.	2026-07-13	In Progress

```
8 rows in set (0.00 sec)

mysql>
mysql> SELECT * FROM Services;
```

ServiceID	ServiceName	Description	Availability
1	Laundry	Clothes washing service	Available
2	Medical	Basic medical assistance	Available
3	Transportation	Campus transportation service	Available
4	Internet	Wi-Fi support service	Available
5	Food	Mess and food services	Available
6	Counselling	Student counselling service	Available
7	Sports	Sports equipment and facilities	Available

```
7 rows in set (0.00 sec)

mysql>
mysql> SELECT * FROM ServiceRequests;
```

ServiceRequestID	StudentID	ServiceID	RequestDate	Status
1	1	1	2026-07-11	Completed
2	2	2	2026-07-12	Requested
3	3	3	2026-07-13	Completed
4	4	4	2026-07-14	In Progress
5	5	5	2026-07-15	Requested
6	6	6	2026-07-16	Completed
7	7	7	2026-07-17	Requested
8	8	1	2026-07-18	In Progress

```
8 rows in set (0.00 sec)

mysql>
mysql> SELECT * FROM Staff;
```

StaffID	Name	Role	Phone	HostelID
1	Ramesh Kumar	Warden Assistant	9876500011	1
2	Sunita Rao	Cleaner	9876500012	1
3	Vijay Sharma	Electrician	9876500013	2
4	Meena Devi	Cook	9876500014	2
5	Mohan Das	Security Guard	9876500015	3
6	Anil Kumar	Maintenance Worker	9876500016	3

```
6 rows in set (0.00 sec)

mysql>
mysql> SELECT * FROM Visitors;
```

VisitorID	StudentID	VisitorName	Relationship	VisitDate	EntryTime	ExitTime
1	1	Amit Kumar	Brother	2026-07-16	10:00:00	12:00:00
2	2	Neha Sharma	Sister	2026-07-17	11:00:00	13:00:00
3	3	Ravi Reddy	Father	2026-07-18	09:30:00	11:30:00
4	4	Pooja Singh	Mother	2026-07-19	14:00:00	16:00:00
5	5	Aakash Mehta	Brother	2026-07-20	15:00:00	17:00:00
6	6	Neeraj Rao	Father	2026-07-21	10:30:00	12:30:00

```
6 rows in set (0.00 sec)
```

The HostelDB database contains 12 tables that manage student accommodation and hostel services. The Students, Hostels, Rooms, and Beds tables handle basic accommodation information, while Allocations connects students with their assigned beds. Payments manages fee records, and Complaints and MaintenanceRequests track student issues. The Services and ServiceRequests tables manage student services, while Staff and Visitors maintain hostel staff and visitor information. Together, these related tables provide a centralized and organized system for hostel management.

Expected Outcomes

Expected Outcomes

Centralized hostel management database

Accurate room and bed availability

Prevention of duplicate allocations

Efficient fee and payment tracking

Complete student allocation history

Organized visitor records

Faster complaint resolution

Efficient maintenance tracking

Accurate occupancy and vacancy reports

Improved data integrity and reliability

Future Implementation

MySQL Database + Python Application

MySQL
Relational Database
Engine

Python
Application Layer

Thank You